VASAVI PRASANNA KURAPATI

AI/ML Engineer | Generative AI, LLM Systems & Applied Machine Learning

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PROFESSIONAL SUMMARY

AI/ML Engineer with 4+ years of applied machine learning experience designing and shipping production ML and LLM systems for banking and fintech. Fine-tunes large language models, engineers real-time fraud and credit-risk pipelines, and owns the full ML lifecycle from data engineering through MLOps and cloud-native delivery (AWS, Azure OpenAI, Docker, Kubernetes). Partners closely with data engineering, risk, and compliance teams to turn research prototypes into reliable, high-throughput services that meet regulatory and business requirements.

CORE TECHNICAL SKILLS

Programming Languages: Python, SQL, PySpark, JavaScript, TypeScript**,** R, C++

Generative AI & LLMs: OpenAI GPT-4, Azure OpenAI, LangChain, LangGraph, LlamaIndex, Hugging Face Transformers, RAG (Retrieval-Augmented Generation), Agentic AI, AI Agents, Prompt Engineering, Prompt Tuning, Fine-Tuning (SFT), RLHF, LoRA, PEFT, Embeddings, Semantic Search, Hybrid Search, Function Calling, Tool Calling, Vector Search

Machine Learning & Deep Learning: Scikit-learn, TensorFlow, PyTorch, Keras, XGBoost, LightGBM, CNN, RNN, LSTM, Transformers, BERT, T5, Llama, Computer Vision, Natural Language Processing (NLP), Feature Engineering, Model Training, Model Evaluation, Model Optimization

Libraries & Frameworks: Pandas, NumPy, SciPy, Matplotlib, Seaborn, OpenCV, FastAPI, Flask, Django, TensorBoard

MLOps & LLMOps: MLflow, Docker, Kubernetes, SageMaker Pipelines, Kubeflow, Apache Airflow, Jenkins, GitHub Actions, CI/CD, Model Deployment, Model Monitoring, Experiment Tracking, Feature Store

Cloud Platforms: AWS (SageMaker, EC2, S3, Lambda, ECS, ECR, IAM, CloudWatch), Microsoft Azure (Azure OpenAI, Azure AI Search, Azure Machine Learning), Google Cloud Platform (GCP)

Data Engineering & Streaming: Apache Spark, PySpark, Apache Kafka, ETL/ELT Pipelines, Data Processing, Data Engineering

Vector Databases & Databases: Pinecone, Weaviate, FAISS, ChromaDB, PostgreSQL, MySQL, MongoDB, Redis

APIs & Backend Development: REST APIs, FastAPI, Flask, Django, gRPC, GraphQL, OAuth2, Microsoft Entra ID

Version Control & DevOps: Git, GitHub, GitHub Actions, Jenkins, Docker, Kubernetes, Terraform

Operating Systems: Linux, Windows, macOS

PROFESSIONAL EXPERIENCE

Principal Financial Group — **Jan 2025 – Present**

AI/ML Engineer — *Wilmington, DE*

- Fine-tuned LLMs (Llama 2, Mistral) via SFT and RLHF with FSDP distributed training, reducing perplexity 7.0→5.0 and lifting downstream task accuracy by 15%.
- Architected real-time fraud-screening pipelines (AWS, Python, PySpark, Kafka) with Docker/Kubernetes anomaly detection microservices, cutting fraud losses by 30% across monitored transaction flows.
- Productionized credit-risk models (XGBoost, Logistic Regression) through MLflow/FastAPI into lending decision engines, reducing loan-approval turnaround by 30%.
- Developed RAG-powered LLM copilots (BERT/T5, Azure OpenAI, vector DBs) adopted by 40+ risk and research analysts, reducing document-research time by ~40%.
- Automated end-to-end ML pipelines (training→validation→deployment) with MLflow/CI-CD, shortening release cycles from ~2 weeks to under 3 days.
- Deployed 20+ containerized ML/LLM models on AWS (Docker/Kubernetes) as real-time gRPC/REST endpoints, sustaining sub-50ms p95 latency.
- Optimized model-serving infrastructure with quantization and request batching, lowering GPU inference costs by 35% while holding latency within SLA.
- Mentored 2 junior data scientists on MLOps and code-review practices, standardizing deployment workflows and reducing onboarding time.

Bytecraft System — **Mar 2020 – May 2023**

Applied Machine Learning Engineer — *Hyderabad, India*

- Developed ML underwriting models (logistic regression, gradient boosting, clustering) for CredPulse, a credit-risk platform serving 15+ Indian NBFCs/fintechs, improving default-prediction accuracy by 12%.
- Engineered credit features (income stability, repayment behavior, DTI bands), reducing false-positive loan rejections by ~18% and expanding approvable applicant pools.
- Packaged model scripts with Docker and built FastAPI inference wrappers, shortening deployment cycles by 35% and standardizing releases.
- Automated data preprocessing and model-monitoring routines in Python, cutting manual reporting effort by 8 hours/week.

KEY PROJECTS

- **RAG-Powered Enterprise Copilot:** LangChain + Azure OpenAI (GPT-4) + Pinecone; sub-2s response over 100K+ documents, 91% retrieval accuracy.
- **Real-Time Fraud Detection:** PySpark/Kafka/XGBoost on AWS SageMaker; 50K+ transactions/min, <50ms latency, 22% fewer false positives.
- **Geospatial ML & Satellite Analytics:** Random Forest crop-yield model on 10GB+ satellite imagery (Geopandas, Scikit-learn), 87% accuracy; automated via Apache Airflow for weekly retraining.
- **Agentic Fraud Investigation Assistant:** Building a LangGraph + Azure OpenAI agent that autonomously orchestrates transaction history, risk profile, and similar-case retrieval tools to generate analyst-ready fraud escalation recommendations.

EDUCATION

University of Delaware — Newark, DE

Master of Science in Data Science — **Aug 2023 – May 2025**

Malla Reddy University — Hyderabad, India

Bachelor of Technology in Computer Science — **Aug 2016 – Sep 2020**

CERTIFICATIONS & PUBLICATIONS

- Generative AI Engineering Essentials with LLMs; AWS Certified Generative AI Developer - In Progress.
- Co-author, "GMIA-NEXT: Next-Generation Global Map of Irrigated Areas," Research Square preprint, 2026 (DOI: 10.21203/rs.3.rs-10085674/v2) — contributed ground-truth data to a 30-author global ML study.